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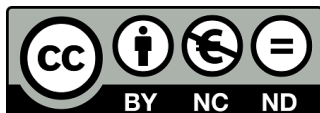
Bachelor Thesis

“High Dimensional Canonical Correlation Analysis: Applications to Financial Data”

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SUMMARY

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1. INTRODUCTION

The rebalancing of portfolios, shifting of investments, and the constant pursuit of outperforming market benchmarks is the day-to-day life of money managers. Whenever an economic slump hits, they turn to defensive stocks for protection, only to rotate back into cyclical stocks at the very first sign of recovery. This balance, despite being under constant scrutiny, is never fully perfected, and it is what we refer to as portfolio optimization.

The theory behind such rebalance is clear: some stocks and sectors perform better during different phases of the economic cycle. While cyclical stocks such as energy and industrials, may give greater returns, they are also sensitive to business cycle fluctuations. In contrast, defensive stocks, usually referring to utilities and consumer staples, provide a “safe haven” during market downturns. The co-movement between these groups therefore contains information about the state of the macroeconomic environment, investor sentiment, and cross-sector capital reallocation. For such reason, understanding the joint dependence structure across sectors helps reveal the extent to which a portfolio is exposed to common macroeconomic risk rather than truly diversified across independent sources of return.

This matters for portfolio optimization because the objective is not simply to choose the best-performing sectors at a single point in time, but to build a portfolio that behaves well across the business cycle. For example, consider two portfolios with the same 60/40 allocation to cyclical and defensive stocks. They may appear identical in terms of weights, but if in one case the two groups have become more highly correlated, diversification is weaker and the defensive allocation provides less protection in adverse states. In that sense, portfolio risk depends not only on portfolio weights, but also on the dependence structure between the assets.

Different investment approaches such as hedging strategies, sector rotation models, and multi-factor asset pricing frameworks all rely on understanding how groups of assets move jointly rather than in isolation. Univariate analyses therefore result insufficient to capture the dependence structure across assets. Determining whether this dependence is low-rank or diffuse (can be explained by a few underlying factors or not) and whether it is dominated by a few directions can serve as a

guideline to inform risk decomposition and factor construction.

Canonical Correlation Analysis (CCA) provides a statistical framework to study such problems. Given two panels of variables, CCA extracts linear combinations within each group that maximize their cross-correlation. The resulting canonical components represent the strongest joint dependence directions between the two sectors. In economic terms, they identify the hidden cross-sector exposures that drive co-movement.

Empirical equity panels, such as those constructed from the S&P 500, are inherently high-dimensional, while time series length is often limited relative to cross-sectional size. In such settings, the classical assumptions underpinning multivariate methods may fail when the number of assets grows with the sample size, estimation errors remain substantial even in large samples, making the sample covariance matrices unfit for inversion.

Recent theoretical advances formalize this phenomenon. In particular, Bykhovskaya and Gorin (2023) establish that when the dimensions of the two panels grow proportionally with the sample size, classical CCA estimators become inconsistent. In these settings, the distribution of canonical correlations converges to a non-degenerate limit (what they call a Wachter-type distribution) instead of settling on their true underlying values. Similarly, the sample directions (canonical vectors) cluster within specific cones around the actual population directions. Only the strongest correlations manage to break away from this general noise, appearing as distinct outliers. The authors identify the point where a correlation becomes strong enough to be detected as a ‘phase transition’.

This result has profound implications. It suggests that in large financial panels, the canonical components obtained by classical CCA may be structurally biased even in large samples. Other common methods used to study cross-sectional dependence also fail to address this issue: PCA ignores cross-panel structure, regression imposes asymmetry, and low-dimensional factor models assume only a few underlying drivers without accounting for errors that arise in high-dimensional settings. In turn, High-Dimensional CCA (HDCCA) provides explicit predictions for how canonical quantities behave when the number of variables is large relative to the sample size.

Despite these advances, it remains unknown whether these high-dimensional asymptotic phenomena meaningfully describe the structure of *real* equity markets. This paper seeks to address this gap in the literature by answering the following question: does HDCCA theory accurately describe

the spectral and geometric structure of cross-sector return dependence in large equity panels? The question focuses on whether canonical correlations between large groups of stocks indeed exhibit the predicted structure and subsequently, whether canonical portfolios are empirically unstable in a manner consistent with the predicted cone geometry.

This contribution is important for three reasons. First, it provides empirical validation of high-dimensional CCA theory in real financial markets. Second, it clarifies whether canonical methods remain informative in large equity universes despite theoretical inconsistency. Third, it offers a structural perspective on cross-sector dependence that bridges random matrix theory and applied asset pricing, thereby informing both risk management and factor modelling practice.

To answer this question, we combine simulation evidence with an empirical application. chapter 2 reviews the main literature, covering classical CCA, its breakdown in high-dimensional settings, and the Bykhovskaya-Gorin framework. chapter 3 presents the Monte Carlo analysis, which assesses the finite-sample relevance of the high-dimensional theoretical predictions under controlled data-generating processes calibrated to realistic financial dimensions. chapter 4 then sets out the empirical methodology, describing the construction of the stock-return panels, the high-dimensional benchmark, and the diagnostic framework used in the data. chapter 5 reports the empirical application to large equity panels. It examines whether the canonical spectrum displays the predicted bulk-outlier structure, whether the leading canonical directions are stable across samples, and whether any apparent dependence survives out of sample in an economically meaningful way. Finally, chapter 6 concludes with a discussion of implications and final remarks.

2. LITERATURE REVIEW

2.1. Classical Canonical Correlation Analysis

Conceptual Overview. Canonical Correlation Analysis (CCA), first introduced by Hotelling (1936), is a classical multivariate statistical method designed to measure the strength of linear association between two random vectors. Let $X_t \in \mathbb{R}^p$ and $Y_t \in \mathbb{R}^q$ denote two zero-mean random vectors observed over $t = 1, \dots, n$. In the context of this paper, these vectors represent panels of cyclical and defensive stock returns, respectively. What makes CCA different than other methods such as PCA is that, rather than studying pairwise correlations between individual variables, CCA seeks linear combinations of each vector that are maximally correlated with one another.

More precisely, CCA seeks vectors $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$ that maximize the correlation between linear projections of the two variable sets:

$$\rho = \max_{a,b} \text{corr}(a^\top X, b^\top Y)$$

Define

$$U = a^\top X, \quad V = b^\top Y,$$

which are referred to as the *canonical variables*. The vectors a and b are known as *canonical directions*, and the resulting correlation between U and V is the *canonical correlation*. Intuitively, CCA identifies the directions in the two variable spaces that exhibit the strongest shared variation. After the first pair of canonical variables is obtained, subsequent pairs are constructed under orthogonality constraints, ensuring that each captures a new independent mode of cross-sectional dependence between the two datasets.

Formally, the first pair of canonical directions (a_1, b_1) solves the optimization problem

$$(a_1, b_1) = \arg \max_{a \in \mathbb{R}^p, b \in \mathbb{R}^q} \frac{a^\top \Sigma_{XY} b}{\sqrt{a^\top \Sigma_{XX} a} \sqrt{b^\top \Sigma_{YY} b}}, \quad (2.1)$$

where

$$\Sigma_{XX} = \mathbb{E}[X_t X_t^\top], \quad \Sigma_{YY} = \mathbb{E}[Y_t Y_t^\top], \quad \Sigma_{XY} = \mathbb{E}[X_t Y_t^\top]$$

denote the covariance matrices within and between the two variable sets. The maximum value of this objective function is the first canonical correlation, denoted ρ_1 . Each pair of canonical variables represents the strongest possible linear relationship between the two datasets. Subsequent canonical pairs (a_k, b_k) are obtained recursively subject to orthogonality constraints

$$a_k^\top \Sigma_{XX} a_j = 0, \quad b_k^\top \Sigma_{YY} b_j = 0, \quad \text{for } j < k. \quad (2.2)$$

which ensure that each successive pair captures a distinct mode of linear dependence between the two datasets.

Eigenvalue Representation. The optimization problem defined in Equation [2.1] seeks to maximize the correlation between linear combinations of the two sets of variables. As demonstrated by Johnson and Wichern, 2007, pp. 539–574 in Chapter 10, solving this constrained maximization through the method of Lagrange multipliers is equivalent to solving a generalized eigenvalue problem. Specifically, the weights are determined by the eigenvectors of the product of the inverse covariance matrices and their cross-covariances, reducing the statistical problem to a computational one of eigenvalue decomposition.

Thus, CCA can be equivalently expressed in terms of an eigenvalue problem. Define the matrix

$$C = \Sigma_{XX}^{-1/2} \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX} \Sigma_{XX}^{-1/2}. \quad (2.3)$$

The squared canonical correlations ρ_k^2 correspond to the eigenvalues of this matrix C , and the canonical directions for X_t correspond to its eigenvectors. Let u_k denote the eigenvector associated with the k -th largest eigenvalue ρ_k^2 (such that $Cu_k = \rho_k^2 u_k$.) Then the canonical directions for X_t are obtained as $a_k = \Sigma_{XX}^{-1/2} u_k$. An analogous representation holds for Y_t .

In empirical applications, population covariance matrices are unknown and replaced by their

sample counterparts:

$$S_{XX} = \frac{1}{n} \sum_{t=1}^n X_t X_t^\top, \quad (2.4)$$

$$S_{YY} = \frac{1}{n} \sum_{t=1}^n Y_t Y_t^\top, \quad (2.5)$$

$$S_{XY} = \frac{1}{n} \sum_{t=1}^n X_t Y_t^\top. \quad (2.6)$$

The sample canonical correlations are therefore the eigenvalues of

$$\widehat{C} = S_{XX}^{-1/2} S_{XY} S_{YY}^{-1} S_{YX} S_{XX}^{-1/2}. \quad (2.7)$$

This representation makes clear that CCA is fundamentally a spectral method, as its statistical properties (both the canonical correlations and directions) depend on the eigenstructure of covariance-based matrix transformations.

Equivalently, CCA can be formulated through the singular value decomposition of $\Sigma_{XX}^{-1/2} \Sigma_{XY} \Sigma_{YY}^{-1/2}$, whose singular values correspond to the canonical correlations.

Statistical Properties Under Classical Asymptotics. Under the classical asymptotic framework in which p and q are fixed while $n \rightarrow \infty$, CCA enjoys well-established statistical guarantees. In particular:

- The sample covariance matrices (S_{XX}, S_{YY}, S_{XY}) are consistent estimators of their population counterparts.
- Sample canonical correlations $\hat{\rho}_k$ converge in probability to the population canonical correlations ρ_k .
- Sample canonical directions converge (up to sign indeterminacy) to their population directions.

Under multivariate normality, likelihood-based inference is possible, and asymptotic distributions for canonical correlations can be derived. In this regime, CCA provides a consistent and efficient method for detecting and estimating cross-dependence between two multivariate systems.

Practical Implications and Limitations. Despite these theoretical guarantees, CCA has historically been less widely used in empirical finance than regression-based methods or principal component analysis (PCA). One reason is computational and statistical fragility as CCA requires inversion of covariance matrices, which becomes unstable when the cross-sectional dimension is large relative to the sample size. In contrast, PCA remains well-defined even when covariance matrices are unstable, and regression frameworks impose a directional structure that simplifies estimation. Moreover, classical asymptotic theory assumes fixed dimension and large sample size, conditions that are rarely satisfied in financial panels with many assets.

In recent times, the increasing availability of large cross-sectional datasets has renewed interest in multivariate techniques capable of capturing joint dependence structures. Unlike PCA, which focuses on variation within a single panel, CCA directly targets cross-panel relationships. In settings where understanding interactions between groups of assets is central, CCA offers a sound framework for capturing shared variation. Most literature has focused on domains such as neuroimaging, genomics, or ecology. More recently, due to the advances in data collection methods, it has been applied in natural language processing and computer vision to align textual and visual representations, enabling cross-modal retrieval and feature learning (Andrew et al., 2013; Gatto & dos Santos, 2017; Sun et al., 2019; Yang et al., 2017, 2021). Despite less frequent application to financial data, some include efforts to study relationships between asset returns and macroeconomic variables, to extract joint factor structures, and to analyze cross-market dependence (Firoozye et al., 2023). Overall, its symmetry provides an advantage in these contexts as CCA does not impose a directional relationship between X_t and Y_t , but instead identifies mutually maximizing linear combinations that capture shared variation.

Nonetheless, these classical consistency results rely critically on the assumption that dimensionality remains fixed as the sample size grows. When p and q increase proportionally with n , the spectral behavior of sample covariance matrices changes fundamentally, and classical asymptotic theory no longer applies. Understanding CCA in such settings requires a high-dimensional asymptotic framework.

2.2. Breakdown in High Dimensions

Dimensionality and Covariance Instability. The classical properties of CCA rely critically on the assumption that the dimensions p and q remain fixed as the sample size n increases. In financial datasets, this is usually not the case as large equity universes often contain hundreds of assets, while the available time series length may be only a few thousand observations or less. In such settings, the ratios

$$\frac{p}{n} \rightarrow c_1 > 0, \quad \frac{q}{n} \rightarrow c_2 > 0$$

are non-negligible.

When p or q is comparable to n , sample covariance matrices become unstable. If $p \geq n$, the matrix S_{XX} is singular and cannot be inverted, rendering classical CCA undefined without regularization. Even when $p < n$ but p/n is not small, eigenvalues of sample covariance matrices exhibit systematic distortion relative to their population counterparts. This phenomenon is well-documented in random matrix theory and leads to biased spectral estimates.

Because CCA is fundamentally a spectral method, relying on inverses and eigenvalues of covariance-based matrices, these distortions directly affect canonical correlations and canonical directions.

Eigenvalue Distortion and Spurious Correlations. In high-dimensional settings, even when the true cross-covariance Σ_{XY} is zero, sample canonical correlations will still report nontrivial correlations. Instead of converging to zero, their empirical distribution converges to a non-degenerate limiting law. This implies that apparent cross-panel dependence may arise purely from dimensionality-induced noise.

More generally, when a finite number of population canonical correlations are present, the empirical canonical correlations do not converge directly to their population values. Instead, they follow a structured bulk–outlier pattern. A large fraction of empirical canonical correlations accumulate within a deterministic interval (the “bulk”), while only sufficiently strong population correlations generate isolated eigenvalues outside this region.

The High-Dimensional CCA Framework. These phenomena are formalized in *High-Dimensional Canonical Correlation Analysis* by Bykhovskaya and Gorin (2023). They consider a regime in which

$$p, q, n \rightarrow \infty \quad \text{with} \quad \frac{p}{n} \rightarrow c_1 \in (0, 1), \quad \frac{q}{n} \rightarrow c_2 \in (0, 1).$$

Under suitable regularity conditions, they show that:

- The empirical distribution of squared canonical correlations converges to a deterministic limit characterized by a Wachter-type distribution.
- Population canonical correlations below a critical threshold are not individually identifiable: their sample counterparts are absorbed into the bulk.
- Only population correlations exceeding a phase transition threshold generate isolated empirical eigenvalues.

This phase transition behavior is analogous to the Baik–Ben Arous–Péché (BBP) phenomenon in random matrix theory. It implies that detectability of cross-panel dependence depends not only on signal strength but also on dimensional ratios c_1 and c_2 .

Inconsistency of Canonical Directions. In such high-dimensional framework, classical CCA estimators of canonical directions are inconsistent. Even when a population canonical correlation exceeds the phase transition threshold and produces an isolated empirical eigenvalue, the associated sample canonical vector does not converge to its population counterpart.

Instead, the sample direction lies within a deterministic cone around the true direction. The limiting cosine of the angle between the sample and population vectors is strictly less than one and depends explicitly on signal strength and dimensional ratios. In other words, estimation error does not vanish asymptotically; it stabilizes at a non-zero level determined by the relative size of the dimensionality and the sample size.

Correction and Benchmark Implications. Importantly, the theory does not only diagnose the inconsistency, it quantifies it. Given observed empirical canonical correlations and known dimen-

sional ratios $(p/n, q/n)$, the framework provides formulas that map sample quantities to implied population parameters. In particular:

- One can infer the implied population canonical correlation from an observed isolated empirical eigenvalue.
- One can compute the predicted cosine between sample and population canonical directions.

These correction formulas provide a way to interpret canonical components despite their formal inconsistency. Rather than treating estimation error as arbitrary, the high-dimensional framework characterizes its magnitude precisely.

In light of this, the Bykhovskaya-Gorin theory serves as the benchmark for our empirical analysis. Instead of replacing classical CCA with alternative regularization schemes, which present a number of limitations, we evaluate whether the structural patterns predicted by high-dimensional asymptotics are visible in real equity panels. In particular, we investigate whether the distribution of empirical canonical correlation distribution exhibits the predicted bulk-outlier structure and if canonical portfolio instability aligns with theoretical cone-angle predictions, then high-dimensional CCA provides a meaningful structural description of cross-sector dependence.

3. MONTE CARLO SIMULATIONS

The Monte Carlo analysis evaluates whether the distortions predicted by high-dimensional canonical correlation analysis are quantitatively relevant at financially plausible dimensional ratios. This chapter therefore motivates the empirical diagnostics formalized in Chapter 4 by showing how the predicted phenomena manifest under controlled data-generating processes. The detailed figures and tables are reported in Appendix section 7.1.

Three findings are central. First, the pure-noise benchmark shows that naive CCA can produce a large leading sample root even under exact independence. Table 7.1 and Figures 7.1–7.2 show that the null leading root is centered around 0.50, whereas the out-of-sample root and split-sample overlaps are essentially zero. This confirms that large in-sample roots cannot be interpreted as evidence of genuine cross-panel structure without an explicit finite-sample benchmark.

Second, the one-spike and threshold experiments confirm the phase-transition logic. The signal-strength sweep in Table 7.2 and Figures 7.3–7.5 shows that weak spikes remain absorbed into the bulk, while the threshold zoom in Table 7.3 and Figures 7.6–7.8 shows that detection improves only gradually around $\rho_c \approx 0.426$. Even above the threshold, geometric recovery and out-of-sample performance improve more slowly than the in-sample spectrum.

Third, the richer designs show that empirical separability depends on how the signal is distributed. In the multiple-spike design, only the strongest components are reliably recovered; see Tables 7.4–7.5 and Figure 7.9. In the low-rank versus diffuse comparison, the low-rank design yields clear outliers and materially higher stability, whereas the diffuse design remains close to the bulk; see Table 7.6 and Figures 7.10–7.11.

Finally, the robustness exercise in Table 7.7 and Figure 7.12 shows that the main conclusions survive moderate covariance misspecification, although heavy tails increase in-sample inflation and weaken directional recovery.

These results motivate the empirical methodology developed in Chapter 4. The simulations show that a large leading sample root is economically meaningful only if it exceeds an appropriate

high-dimensional or finite-sample benchmark and is accompanied by directional stability and out-of-sample persistence. They also show that the presence of several nonzero roots is not, by itself, evidence of a strongly identified low-rank dependence structure, since diffuse weak links can generate a broad spectrum without producing robust canonical directions. For that reason, the empirical methodology will evaluate the data jointly through spectral magnitude, benchmark comparison, eigengap separation, vector and subspace stability, and held-out performance, rather than relying on in-sample fit alone.

4. METHODOLOGY

4.1. Objective and empirical strategy

This chapter explains how the empirical analysis is carried out. The goal is to assess whether HDCCA provides a useful description of the dependence between cyclical and defensive stock returns. To do this, classical CCA is applied to rolling windows of stock returns and then the results are evaluated using high-dimensional and finite-sample benchmarks. The analysis asks three main questions. First, are the leading canonical roots larger than would be expected under noise? Second, are the estimated canonical portfolios stable across samples? Third, do they remain informative out of sample?

The purpose is therefore to evaluate whether the distortions predicted for classical CCA in high dimensions are visible in the data.

This approach is deliberately conservative. In high dimensions, large sample roots do not necessarily imply that the corresponding canonical vectors are reliably estimated. For that reason, the empirical analysis does not rely on in-sample fit alone. Instead, it evaluates spectral strength, stability, and out-of-sample performance jointly.

4.2. Empirical setting

The empirical application uses weekly stock returns constructed from CRSP data. The stocks are divided into two balanced panels: cyclical stocks and defensive stocks. The classification is based on FF12 industry groups derived from SIC codes. This partition is economically motivated by the business-cycle distinction and mirrors Bykhovskaya and Gorin's own empirical illustration. Cyclical stocks are defined as firms in durable goods, manufacturing, energy, chemicals, and retail or services, while defensive stocks are defined as firms in non-durables, health, telecom, and utilities. Further data details and sample composition are reported in Chapter 5.

Let $X_t \in \mathbb{R}^p$ denote the vector of cyclical returns and $Y_t \in \mathbb{R}^q$ the vector of defensive returns

in a given week. For each rolling window, classical CCA is used to estimate the sample canonical correlations and the associated canonical vectors.

4.3. Empirical diagnostics

The empirical results are interpreted using a simple rule. Evidence for a strong and economically meaningful low-rank structure requires three conditions at the same time: the leading root must lie clearly above the bulk and null benchmarks, the associated vectors or subspaces must be stable, and the dependence must persist out of sample. If these conditions are not met, a large in-sample canonical root is not treated as evidence of a reliably recoverable canonical portfolio. To evaluate these conditions, six diagnostics are computed for each window.

Under classical low-dimensional asymptotics, sample canonical roots and vectors are consistent. In high dimensions, this is no longer true. When the cross-sectional dimension is large relative to the time-series length, the sample spectrum is distorted, weak signals can be absorbed into the bulk, and even strong outliers may be associated with unstable sample vectors. For this reason, the empirical analysis uses HDCCA as the main benchmark. In each window, the observed canonical roots are compared with the HDCCA bulk edge λ_+ and with a permutation-based null. This first empirical test shows whether the leading root is larger than would be expected under high-dimensional noise. In the permutation null, the weekly time ordering of one panel is randomly permuted relative to the other within the same window, while the observations inside each panel are otherwise left unchanged. This preserves the marginal time-series and cross-sectional structure of each panel but breaks the cross-panel alignment that would generate genuine canonical dependence.

Additionally, the number of canonical roots above the bulk edge in each window is also used as a diagnostic. This second factor helps distinguish a single dominant factor from broader multi-direction dependence.

Roots above this threshold may indicate genuine cross-panel dependence, but this alone is not enough. The corresponding vectors must also be stable. To measure such directional stability each window is split into two halves and the leading canonical vectors across the two subsamples are compared. Low overlap would indicate that the leading canonical portfolio is unstable.

This third measure is complemented by top- k subspace stability, which evaluates the overlap of the leading k -dimensional subspaces across subsamples. This is important because even when the first vector is unstable, the leading subspace may still be well-defined if there are several closely spaced roots. In other words, instability of the first vector may reflect either genuine noise or simple rotation within a weakly separated leading subspace.

The fifth diagnostic is eigengap separation: $\lambda_1 - \lambda_2$, $\lambda_2 - \lambda_3$, and $\lambda_3 - \lambda_4$. These show whether the leading root is sufficiently separated to be interpreted as an individual component. Such step is essential as, when eigengaps are too small, the Bykhovskaya-Gorin inversion formulas cannot be applied credibly.

Finally, out-of-sample performance is evaluated: canonical weights estimated in one part of the data are applied to the other part, and the held-out leading squared canonical correlation is recorded. This shows whether the apparent dependence survives out of sample.

By relying on these six diagnostics jointly, the empirical analysis can evaluate whether large in-sample canonical roots are accompanied by stable and economically meaningful canonical portfolios, or whether they are more consistent with high-dimensional noise and instability.

4.4. Rolling-window design

The baseline specification uses rolling windows of a fixed length of 260 weeks and fixed step size of 4 weeks, yielding 144 windows over the sample. This design allows the canonical spectrum to vary over time while keeping the empirical setting explicitly high-dimensional. For each window, the six diagnostics are computed and is treated as a credible leading-root case only if the first root is clearly separated from the second one.

4.5. Robustness

The main results are checked using alternative window lengths and subsamples. Shorter and longer windows test sensitivity to the time-series dimension, while pre-2020 and post-2020 subsamples test whether the results depend on a particular market period.

5. EMPIRICAL APPLICATION TO STOCK RETURN PANELS

5.1. Data description

The HDCCA framework is applied to a large weekly equity panel constructed from CRSP daily stock data. The final sample contains 832 weekly observations running from 8 January 2010 to 12 December 2025, together with two balanced cross-sections of 80 stocks each: 80 cyclical names and 80 defensive names. In the final selected panel, the cyclical group is composed mainly of retail/services, manufacturing, energy, and chemicals, while the defensive group is concentrated in health, utilities, non-durables, and telecom. This composition guarantees that the two panels represent distinct sectors of the economy and are economically meaningful, as they are expected to differ in their sensitivity to business-cycle conditions.

The central empirical question is whether this panel contains a stable and economically usable leading canonical portfolio, or only strong spectral evidence of cross-panel dependence. As the results below show, the data support the latter much more strongly than the former.

5.2. Main result

Table 6.1 summarizes the baseline 260-week rolling specification. These windows are stepped every four weeks, resulting in 144 rolling windows. With $p=q=80$ and $n=260$, the dimensional ratios are in the high-dimensional region, allowing the use of HDCCA.

The leading empirical root is large, with a mean value of 0.9731 across windows, compared with a mean HDCCA bulk edge of 0.8521. The leading root exceeds the bulk edge in every window, and the mean number of roots above the bulk is 4.17. This effect is strong enough to determine that, at the level of the sample spectrum, cyclical and defensive returns exhibit persistent and meaningful cross-panel dependence.

Already at this stage, the right interpretation is asymmetric: the data show strong spectral dependence, but not yet a stable or economically usable leading canonical portfolio. The rest of

the chapter is devoted to clarifying that distinction.

This conclusion is strengthened by the comparison with the permutation null. Figure 6.1 shows that the leading root is consistently too large to be explained either by the HDCCA bulk edge or by a finite-sample null that destroys cross-panel dependence while preserving the marginal structure of each panel. Figure 6.2 further shows that the evidence is not confined to a single isolated component: multiple roots lie above the bulk in most windows. In that sense, the data reject the interpretation that the observed canonical dependence is merely dimensionality-induced noise.

However, the geometry tells a very different story. The estimated first canonical direction is not reproducible across halves of the sample. The mean split-sample squared cosine overlap of the first vector is only 0.0219 on the cyclical side and 0.0329 on the defensive side, as reported in Table 6.1 and visualized in Figure 6.3. These values are extremely low and imply that the first canonical portfolio extracted in one half of the data contains almost no directional information about the first canonical portfolio extracted in the other half.

The same instability appears out of sample. The mean held-out leading squared root is only 0.0128, which is tiny relative to the mean in-sample value of 0.9731. Such striking comparison can be seen in Figure 6.4: while the in-sample leading root stays close to 0.96 through most windows, the held-out root mostly remains near zero, with only temporary spikes. This is exactly the gap highlighted by the Monte Carlo analysis. A large in-sample canonical root is not enough to be economically meaningful, the corresponding dependence must also survive out of sample.

The overall implication is that the leading empirical root is real as a spectral phenomenon, but the leading empirical vector is not recoverable as a stable economic portfolio. In other words, the spectrum indicates nontrivial cross-sector dependence, but the first canonical portfolio does not behave like a reliable low-rank factor.

5.3. Top-k subspace stability and the rotation problem

The weak split-sample stability of the first vector raises an important ambiguity. A low top-1 overlap does not by itself prove that the leading dependence structure is meaningless. It may instead reflect a rotation problem: if the top few roots are close to each other, different samples may estimate

different linear combinations inside essentially the same low-dimensional subspace.

To address this possibility, the analysis examines top- k subspace overlap for $k = 1, 2, 3, 5$. Figure 6.5 shows that the top- k subspaces are somewhat more stable than the top-1 vectors, suggesting that part of the first-vector instability may be due to rotation among the leading directions rather than pure noise at a single rank. Nonetheless, the gains are modest and the top-5 subspace overlap remains low, at 0.0958 on the cyclical side and 0.1297 on the defensive side (see Table 6.1).

As the leading dependence space is only weakly identified despite allowing for rotational ambiguity, it can be concluded that the evidence is more consistent with clustered or diffuse multi-direction dependence than with a sharply separated low-rank structure.

5.4. Eigengaps and the failure of credible inversion

The eigengap diagnostics reinforce this interpretation. Figure 6.6 shows that the gaps between the leading roots are generally too small to support credible leading-root inversion. In the baseline run, there are zero credible leading windows out of 144, as summarized in Table 6.1. This means that no rolling window satisfies the separation rule required to safely apply the Bykhovskaya-Gorin outlier formulas on a window-by-window basis.

This finding has two consequences. First, it explains why the first-vector overlaps are so low: when λ_1 and λ_2 are not well separated, the estimated leading eigenvector can rotate substantially across samples even if the broader leading space contains real dependence. Second, it implies that the thesis should not claim successful empirical recovery of implied population signal strength from rolling outliers. The identification conditions required for that exercise are simply not met.

Accordingly, the empirical application does not provide evidence of a clean window-by-window phase-transition recovery exercise. Instead, it provides a different and more important result: in this stock-return panel, the leading roots are spectrally strong but geometrically fragile. That is itself a central empirical confirmation of the HDCCA perspective.

5.5. Interpreting the spectrum

Taken together, the spectral and geometric evidence points to a clear interpretation. Cross-sector dependence between cyclical and defensive stock returns is genuine, strong, and persistent. It is not plausibly attributable to noise alone. At the same time, the leading empirical canonical portfolio is not a stable object. The estimated first direction changes materially across halves of the sample and fails to transport out of sample.

This pattern is difficult to reconcile with a clean one-factor or sharply separated low-rank dependence model. If the dependence were dominated by one robust latent factor, one would expect not only a large first outlier, but also reproducible leading weights and nontrivial held-out canonical strength. Instead, the evidence suggests a more diffuse structure in which several nearby directions contribute to cross-panel dependence, but no single one is stably estimated as the dominant portfolio. That reading also matches the Monte Carlo comparison between concentrated low-rank and diffuse dependence designs, where broad nonzero spectra can coexist with weak directional recovery.

5.6. Robustness

The robustness exercises show that the main conclusion is not driven by the baseline window length or sample period as seen in Table 6.2 and Figure 6.7.

With shorter 208-week windows, the leading root remains an outlier in every window, but vector stability, top-5 subspace overlap, and held-out performance all remain very weak, and there are again no credible leading windows. With longer 364-week windows, spectral separation becomes stronger and the mean held-out root rises materially to 0.3385, well above the baseline value of 0.0128. This is economically important and deserves interpretation. Longer windows evidently allow a larger part of the broad cross-panel dependence to survive out of sample, which is consistent with lower estimation noise and with more persistent low-frequency co-movement being captured over longer horizons. However, this improvement does not solve the main identification problem. First-vector overlap remains very low, top-5 overlap stays near 0.10, and the number of credible leading windows is still zero. In other words, longer windows strengthen evidence that

the dependence is real at the spectral level, but they still do not deliver a stable leading canonical portfolio that could be interpreted as a clean, well-identified factor.

The subsample analysis leads to the same conclusion. In both the pre-2020 and post-2020 samples, the leading root remains an outlier throughout, but overlap measures remain low, held-out roots remain small, and no credible leading windows emerge.

Overall, the robustness results confirm the baseline message: the spectrum consistently indicates cross-sector dependence, but the leading canonical portfolio is not a well-identified economic direction under reasonable alternative specifications.

6. DISCUSSION AND CONCLUSION

This paper studied whether HDCCA provides a useful description of cross-sector dependence in large stock-return panels. The main result is that it does. As shown in the empirical application, the canonical spectrum displays strong and persistent dependence between cyclical and defensive stocks, but the leading canonical portfolio is not recovered as a stable economic direction.

Put differently, the empirical application finds real spectral dependence, but not a stable, usable leading canonical portfolio. That distinction is central for the interpretation of canonical methods in high dimensions: the spectrum can contain genuine information even when the leading estimated direction is too fragile to support a clean portfolio or factor reading.

This finding helps clarify when classical CCA fails in high dimensions. A large in-sample canonical root is not enough to claim that a meaningful leading factor has been identified. In this setting, classical CCA can detect strong spectral dependence while still delivering unstable canonical vectors. That is exactly what the empirical results show: the leading root is consistently above the HDCCA bulk benchmark and the permutation null, but vector stability is very low, out-of-sample performance is weak, and the leading root is not sufficiently separated from nearby roots for credible inversion. Thus, HDCCA is useful not because it guarantees recovery of a stable canonical portfolio, but because it explains why large canonical roots in high-dimensional data must be interpreted cautiously.

The practical implication is straightforward. In large financial panels, classical CCA should not be interpreted using in-sample fit alone. For portfolio construction, risk management, or factor interpretation, large canonical roots are only economically meaningful if they are also stable across samples and persist out of sample. Otherwise, the estimated canonical portfolio may reflect high-dimensional estimation error rather than a reliable source of common risk.

More broadly, the results suggest that cross-sector dependence in equity markets may be real but more diffuse than a simple low-rank model would imply. The evidence is therefore more consistent with several weakly separated directions than with one clean and stable leading canonical factor.

There are some limitations to the study as the empirical analysis focuses on one panel design, one frequency, and classical CCA interpreted through the HDCCA benchmark. It does not compare alternative regularized estimators in detail, nor does it study other asset partitions or richer time-series structures. These limitations point naturally to future research. Extension would include a comparison of classical CCA with regularized or sparse alternatives, as well as to study whether economically meaningful information survives more clearly at the subspace level than at the level of the first canonical vector.

BIBLIOGRAPHY

- Andrew, G., Arora, R., Bilmes, J., & Livescu, K. (2013). Deep canonical correlation analysis. In S. Dasgupta & D. McAllester (Eds.), *Proceedings of the 30th international conference on machine learning* (pp. 1247–1255, Vol. 28). PMLR. <https://proceedings.mlr.press/v28/andrew13.html>
- Bykhovskaya, A., & Gorin, V. (2023). High-dimensional canonical correlation analysis. *arXiv preprint arXiv:2306.16393*.
- Firoozye, N., Tan, Q., & Zohren, S. (2023). Canonical portfolios: Optimal asset and signal combination. *Journal of Banking & Finance*, 154, 106952. <https://doi.org/10.1016/j.jbankfin.2023.106952>
- Gatto, B. B., & dos Santos, E. M. (2017). *Discriminative canonical correlation analysis network for image classification*. <https://doi.org/10.1109/ICIP.2017.8297131>
- Hotelling, H. (1936). Relations between two sets of variates. *Biometrika*, 28(3/4), 321–377.
- Johnson, R. A., & Wichern, D. W. (2007). *Applied multivariate statistical analysis* (6th). Pearson Prentice Hall.
- Sun, Z., Sarma, P. K., Sethares, W. A., & Bucy, E. P. (2019). Multi-modal sentiment analysis using deep canonical correlation analysis. *CoRR*, abs/1907.08696. <http://arxiv.org/abs/1907.08696>
- Yang, X., Liu, W., Liu, W., & Tao, D. (2021). A survey on canonical correlation analysis. *IEEE Transactions on Knowledge and Data Engineering*, 33(6), 2349–2368. <https://doi.org/10.1109/TKDE.2019.2958342>
- Yang, X., Liu, W., Tao, D., & Cheng, J. (2017). Canonical correlation analysis networks for two-view image recognition. *Information Sciences*, 385-386, 338–352. <https://doi.org/https://doi.org/10.1016/j.ins.2017.01.011>

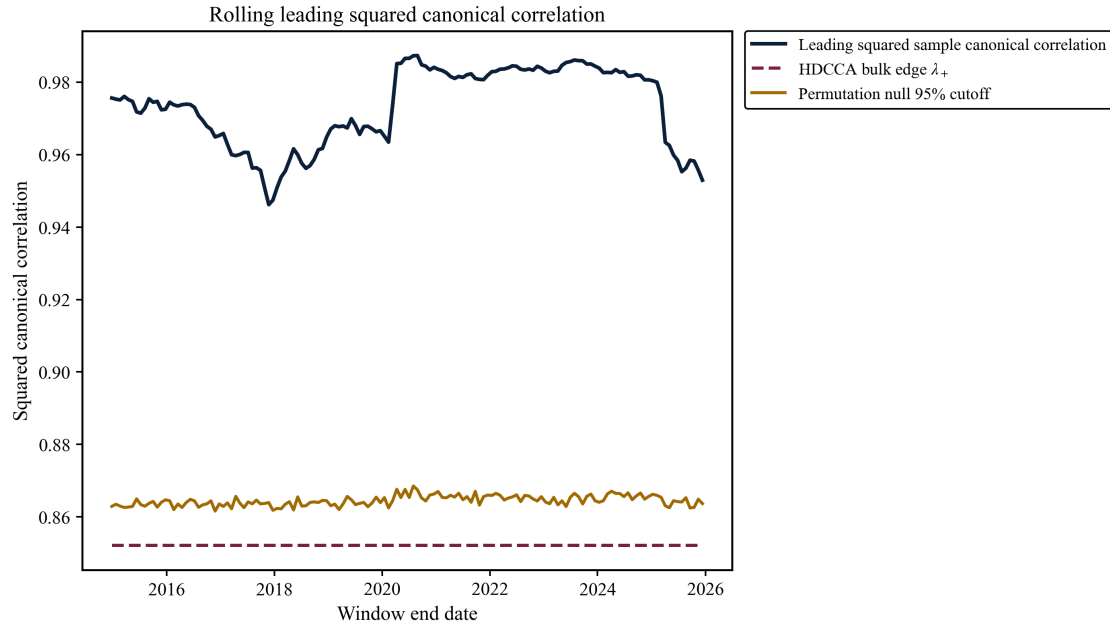
Metric	Value
Rolling windows	144
Window length (weeks)	260
Panel dimensions (p, q)	(80, 80)
Mean leading squared root $\hat{\lambda}_1$	0.9731
Mean HDCCA bulk edge λ_+	0.8521
Share with $\hat{\lambda}_1 > \lambda_+$	100.0%
Mean number of outliers above bulk	4.17
Mean split stability, cyclical u_1	0.0219
Mean split stability, defensive v_1	0.0329
Mean held-out leading squared root	0.0128
Credible leading windows	0 / 144
Mean top-5 overlap, cyclical side	0.0958
Mean top-5 overlap, defensive side	0.1297

Table 6.1

Baseline 260-week rolling summary for the CRSP empirical application.

Figure 6.1

Baseline rolling leading squared canonical root against the HDCCA bulk edge and the permutation null benchmark.



Specification	Windows	Mean $\hat{\lambda}_1$	Mean λ_+	Mean held-out root	Mean top-5 cyclical	Credible share
Baseline 260w	144	0.9731	0.8521	0.0128	0.0958	0.0000
Short 208w	157	0.9839	0.9467	0.0205	0.0991	0.0000
Long 364w	118	0.9612	0.6859	0.3385	0.0995	0.0000
Pre-2020 260w	66	0.9657	0.8521	0.0108	0.0953	0.0000
Post-2020 260w	13	0.9638	0.8521	0.0171	0.1031	0.0000

Table 6.2

Robustness summary across alternative rolling-window lengths and subsamples.

Figure 6.2

Baseline rolling count of squared canonical roots above the HDCCA bulk edge.

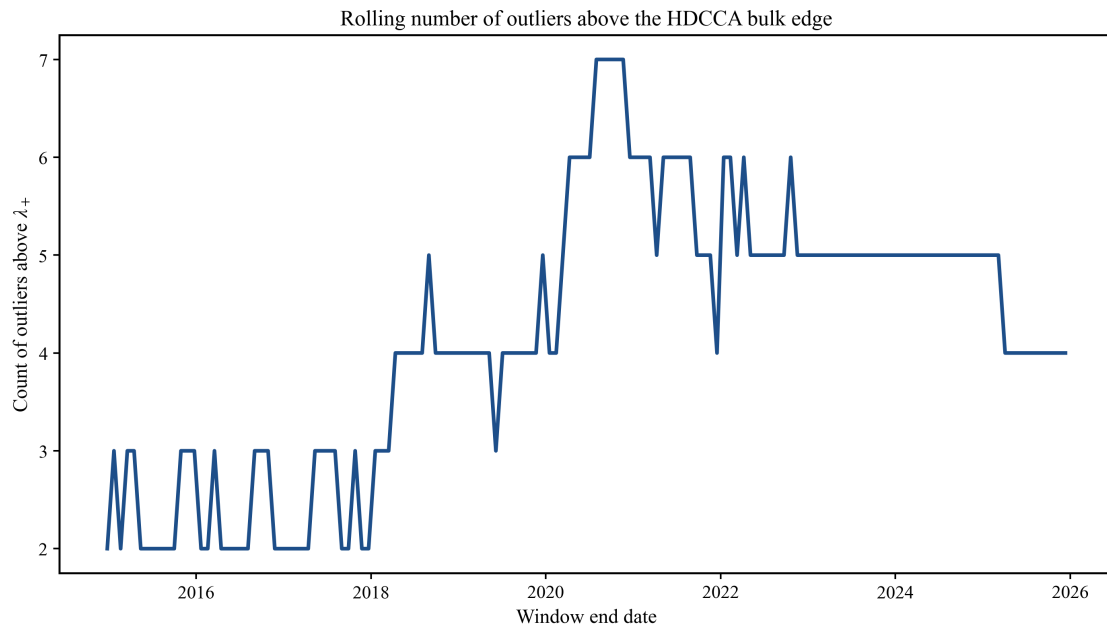


Figure 6.3

Baseline split-sample stability of the first canonical direction on the cyclical and defensive sides.

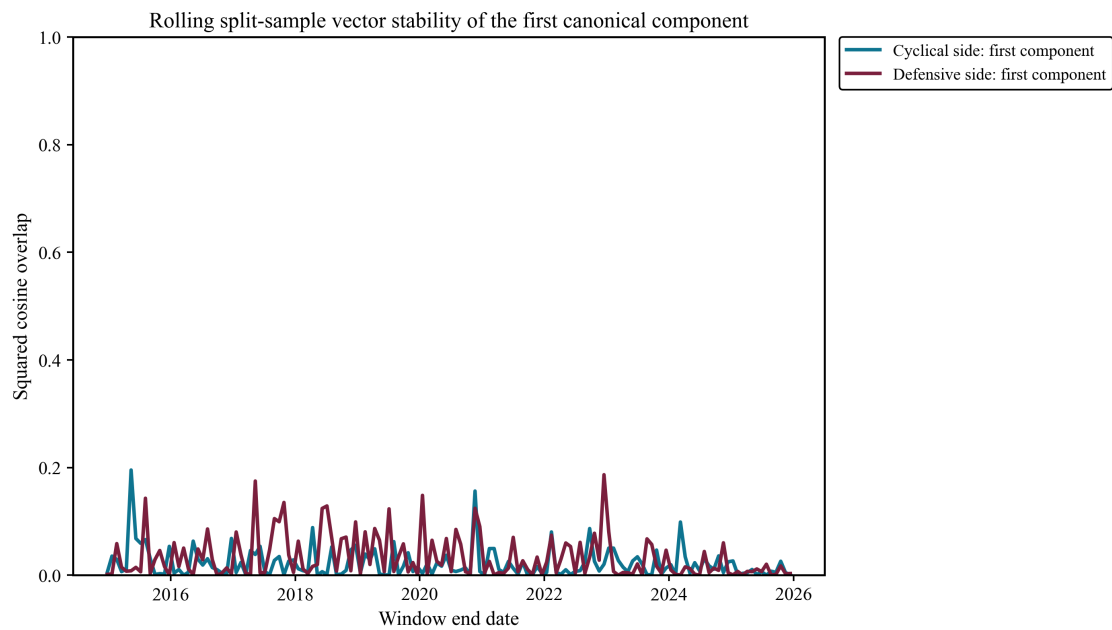


Figure 6.4

Baseline comparison between the in-sample leading squared root and the held-out leading squared root.

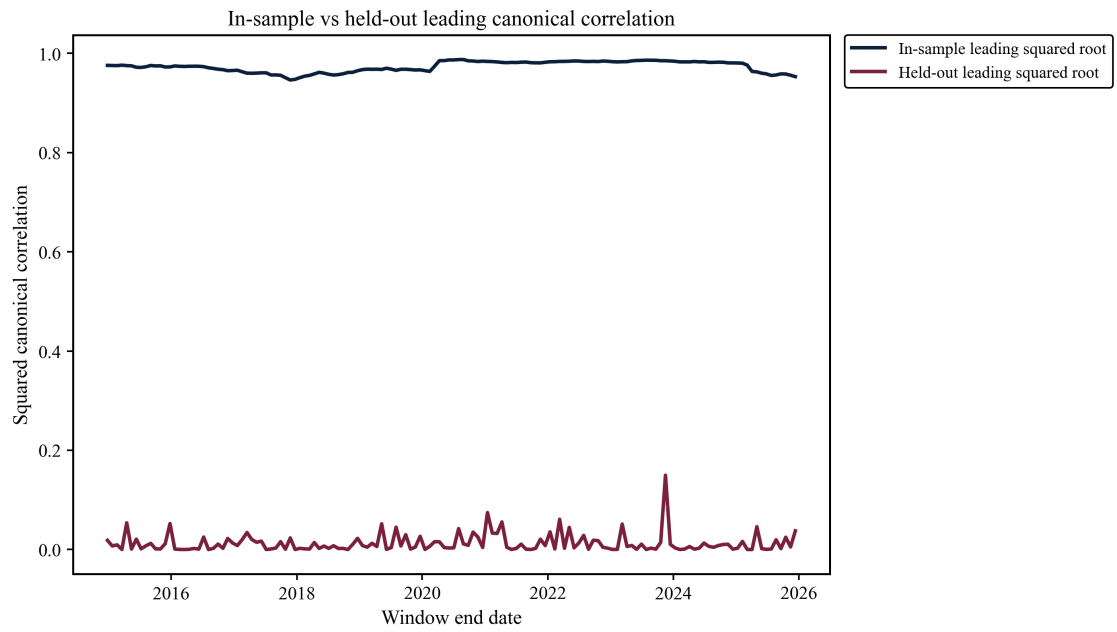


Figure 6.5

Baseline top-k subspace overlap for $k = 1, 2, 3$, and 5 on the cyclical and defensive sides.

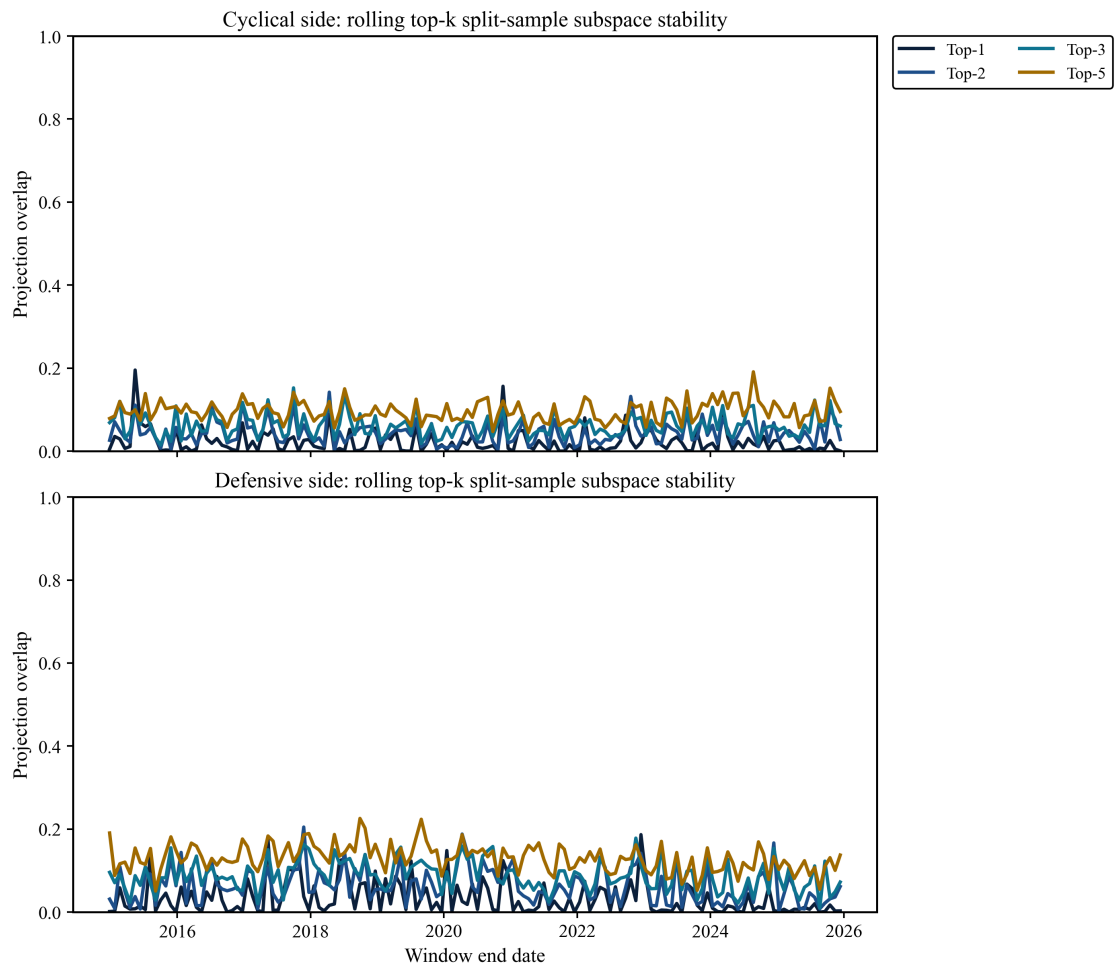
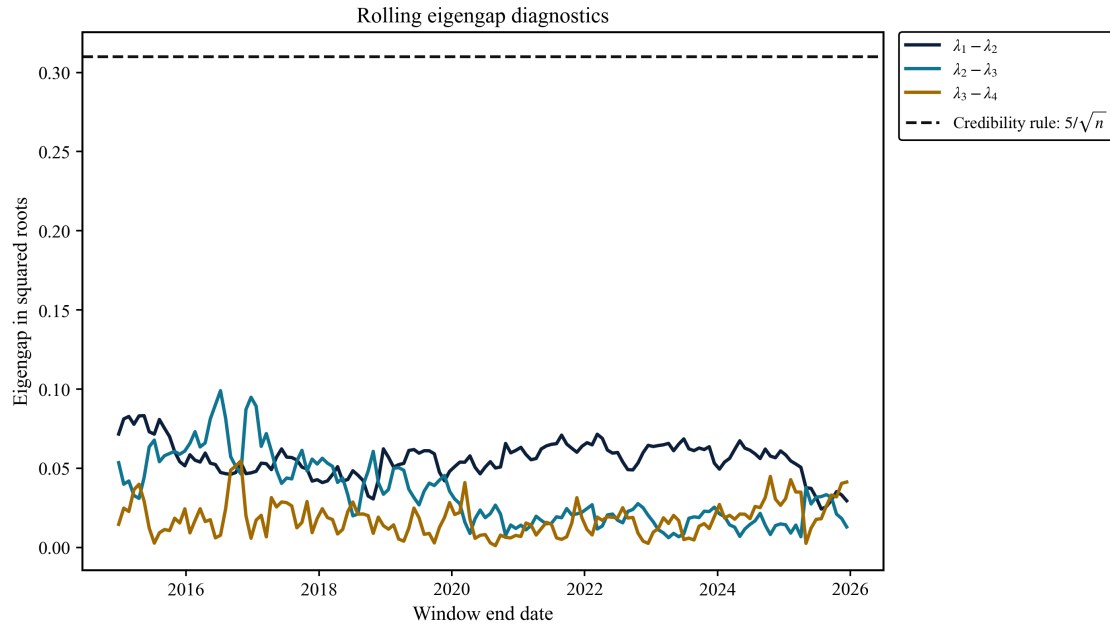
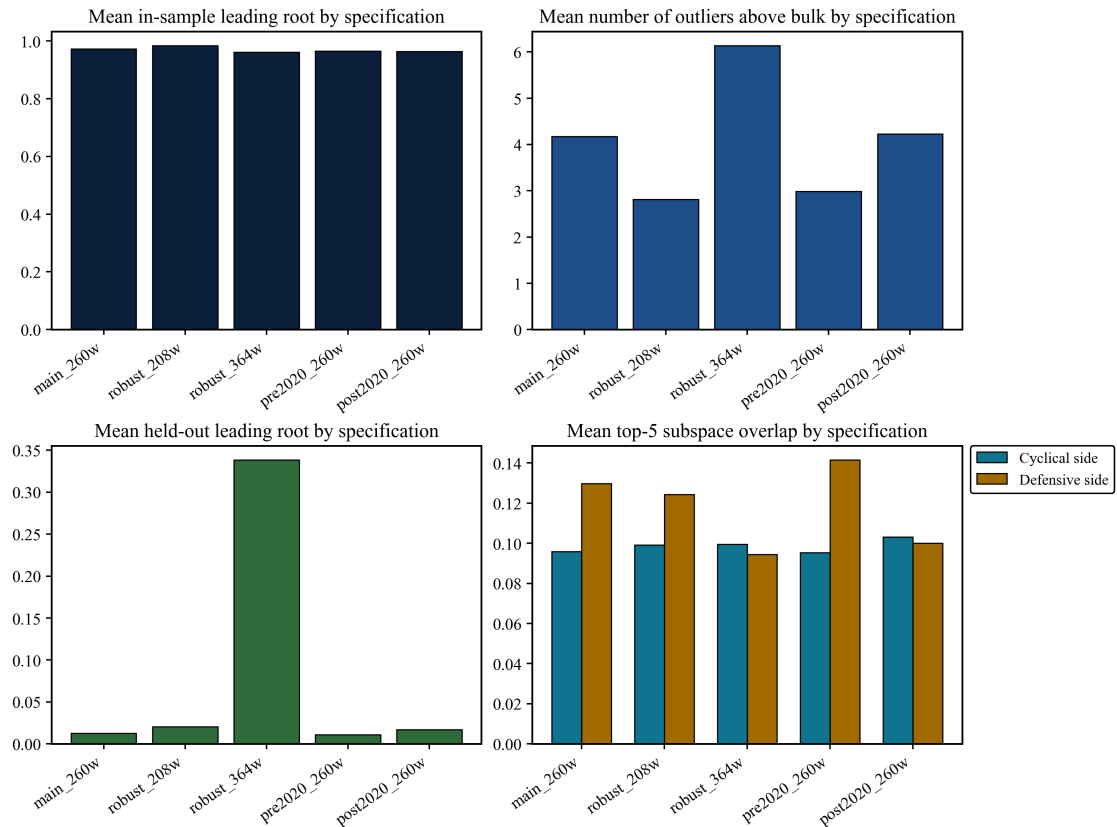


Figure 6.6

Baseline eigengap diagnostics for the first few rolling canonical roots.

**Figure 6.7**

Cross-specification robustness comparison for spectral strength, stability, and held-out dependence.



7. APPENDIX

7.1. APPENDIX 1: Monte Carlo Full Results

This appendix reports the full Monte Carlo evidence underlying the main results. The discussion below summarizes the main patterns in the simulations, while the complete tables and figures are collected afterward on a separate page.

Experiment 1: Pure-Noise Benchmark

Experiment 1 provides the null benchmark for the entire Monte Carlo study. When the two panels are truly independent, the leading sample canonical root is nevertheless large in finite samples: the mean of $\hat{\lambda}_1$ is 0.5047, with a standard deviation of 0.0156. The 95% and 99% null cutoffs are 0.5312 and 0.5454, respectively. This shows that, in a high-dimensional setting, naive CCA can generate apparently strong in-sample dependence even when the true cross-panel correlation is exactly zero; see Table 7.1 and Figures 7.1–7.2.

The null design also reveals that this apparent signal is entirely unstable. The mean out-of-sample leading root is only 0.0039, while the split-sample squared cosine overlaps are approximately 0.0124 for the X -side vectors and 0.0124 for the Y -side vectors. In other words, the canonical directions extracted under the null do not replicate across samples. Taken together, these results imply that a large leading sample root is not, by itself, evidence of meaningful dependence. This experiment therefore provides the finite-sample reference distribution used later to distinguish genuine outliers from noise-generated spikes.

Experiment 2: One-Spike Signal-Strength Sweep

Experiment 2 studies how the leading canonical root reacts when a single true population spike is introduced and its strength is varied. The theoretical high-dimensional threshold is approximately $\rho_c \approx 0.426$, so the key question is whether the sample spectrum begins to separate from the null

benchmark before that point. The results indicate that, throughout the weak-signal region, the leading sample root remains very close to its null value. As ρ increases from 0 to 0.345, the mean of $\hat{\lambda}_1$ rises only from 0.5047 to 0.5098, while the second sample root remains essentially unchanged, moving only from 0.4793 to 0.4840. Hence, for subcritical or near-subcritical signal strengths, the sample spectrum is still largely dominated by the high-dimensional noise bulk; see Table 7.2 and Figure 7.3.

The instability diagnostics reinforce this conclusion. Over the same range, the mean out-of-sample leading root stays near zero, fluctuating around 0.004, which is negligible relative to the in-sample value of about 0.50. Likewise, the estimated canonical vectors remain poorly recovered: the average angles are still extremely large, falling only from about 85° to 78° , and the squared cosine overlaps rise only modestly from roughly 0.01 to 0.06. These findings imply that weak population dependence is absorbed into the bulk rather than emerging as a clean, interpretable outlier. Thus, before the signal reaches the threshold region, naive CCA substantially overstates the strength of dependence in sample while delivering almost no out-of-sample persistence or reliable vector recovery; see Figures 7.4 and 7.5.

Experiment 3: Threshold Zoom

Experiment 3 focuses on the neighborhood of the theoretical high-dimensional threshold, which in this calibration is approximately $\rho_c \approx 0.426$. The main result is that detection improves only gradually through this region rather than appearing as a sharp finite-sample jump. When $\rho = 0.306$, the empirical detection probability is only 0.0148; at $\rho = 0.346$ it remains 0.0190; at $\rho = 0.386$ it is still only 0.0326. Even exactly at the theoretical threshold, the detection probability is just 0.0598. Clearer separation only emerges above the threshold: detection rises to 0.1370 at $\rho = 0.466$, to 0.3324 at $\rho = 0.506$, and to 0.6130 at $\rho = 0.546$; see Table 7.3 and Figure 7.6.

The other diagnostics show that reliable recovery lags behind in-sample spectral separation. The mean leading sample root increases from 0.5087 below the threshold to 0.5531 at the strongest value considered, but the mean out-of-sample root remains extremely small, rising only from 0.0038 to 0.0058 over the same range. At the same time, the canonical directions become gradually more stable: the average angle on the X side falls from about 80° at $\rho = 0.306$ to about 47° at $\rho = 0.546$,

while the squared cosine overlap rises from 0.0439 to 0.4674. Thus, the threshold region should be interpreted as a transition zone rather than a knife-edge boundary. Above ρ_c , outliers become more likely, but vector recovery and out-of-sample persistence still improve only progressively; see Figures 7.7 and 7.8.

Experiment 4: Multiple-Spike Low-Rank Design

The following experiment studies a multi-spike low-rank design. Specifically, three spikes with population canonical correlations (0.80, 0.55, 0.30) are considered. The results show strong heterogeneity across components. On average, the number of detected outliers is 1.5785, which implies that finite samples typically separate only the strongest part of the low-rank structure. The mean leading three sample roots are 0.7544, 0.5515, and 0.4934, respectively, so the spectrum remains clearly ordered, but only the first two components are meaningfully distinguishable from the noise benchmark; see Table 7.4 and Figure 7.9.

The recovery metrics confirm this ranking. The first component is estimated well: its mean angle on the X side is only 20.84° , with mean squared cosine overlap 0.8725. The second component is recovered only moderately well, with mean angle 47.05° and mean squared cosine overlap 0.4692. By contrast, the third component is effectively lost in the bulk: its mean angle is 80.45° and its mean squared cosine overlap is only 0.0407. The average out-of-sample leading root is 0.1881, and the mean split-sample squared cosine overlaps are about 0.272 on both sides, again indicating that only part of the latent low-rank structure is stably recovered. Overall, the experiment supports the HDCCA prediction that in a multi-spike design only sufficiently strong spikes emerge as reliable empirical outliers, while weaker ones remain statistically present but practically unrecoverable; see Table 7.5.

Experiment 5: Diffuse Versus Low-Rank Dependence

Experiment 5 is reported in comparative tabular and graphical form in Table 7.6 and Figures 7.10–7.11. Its purpose is to contrast the empirical scree behavior under a concentrated low-rank dependence structure and a diffuse alternative.

Experiment 6: Finance-Style Robustness Checks

Experiment 6 examines whether the main qualitative conclusions survive under more realistic data-generating mechanisms. Across all three settings, the leading root is always detected, with empirical detection probability equal to 1. This indicates that a population signal of size $\rho = 0.70$ is strong enough to remain visible even when the benchmark assumptions are relaxed. Under the Gaussian identity design, the mean leading sample root is 0.6578; under Gaussian spiked within-panel covariance it is 0.6577; and under Student- t innovations it rises to 0.7135. Thus, heavy tails generate noticeably stronger in-sample inflation, whereas moderate within-panel covariance heterogeneity leaves the leading root almost unchanged relative to the benchmark; see Table 7.7 and Figure 7.12.

The recovery and stability measures are more informative than detection alone. In the Gaussian identity and Gaussian spiked-covariance cases, the mean out-of-sample leading root is about 0.04, the mean angle is about 28.5° , and the mean squared cosine overlap is about 0.77 on both sides. These two settings therefore deliver very similar qualitative conclusions: the spike is detectable and the associated vectors are recovered reasonably well. Under Student- t innovations, however, recovery is visibly weaker. The mean out-of-sample leading root falls to 0.0314, the mean angle increases to about 35.2° , and the mean squared cosine overlap declines to about 0.667. The split-sample overlaps also fall from roughly 0.06 in the Gaussian settings to about 0.04 under heavy tails. Therefore, the main HDCCA conclusions are robust to mild covariance misspecification, but heavy tails make the signal look stronger in sample than it is out of sample and reduce the reliability of vector recovery.

Experiment 1: Pure-Noise Benchmark

Metric	Value
Mean $\hat{\lambda}_1$	0.5047
Standard deviation of $\hat{\lambda}_1$	0.0156
95% null cutoff for $\hat{\lambda}_1$	0.5312
99% null cutoff for $\hat{\lambda}_1$	0.5454
Mean out-of-sample $\hat{\lambda}_1$	0.0039
Mean split stability, X	0.0124
Mean split stability, Y	0.0124

Table 7.1

Experiment 1 summary statistics under the pure-noise benchmark.

Figure 7.1

Experiment 1 mean null scree plot.

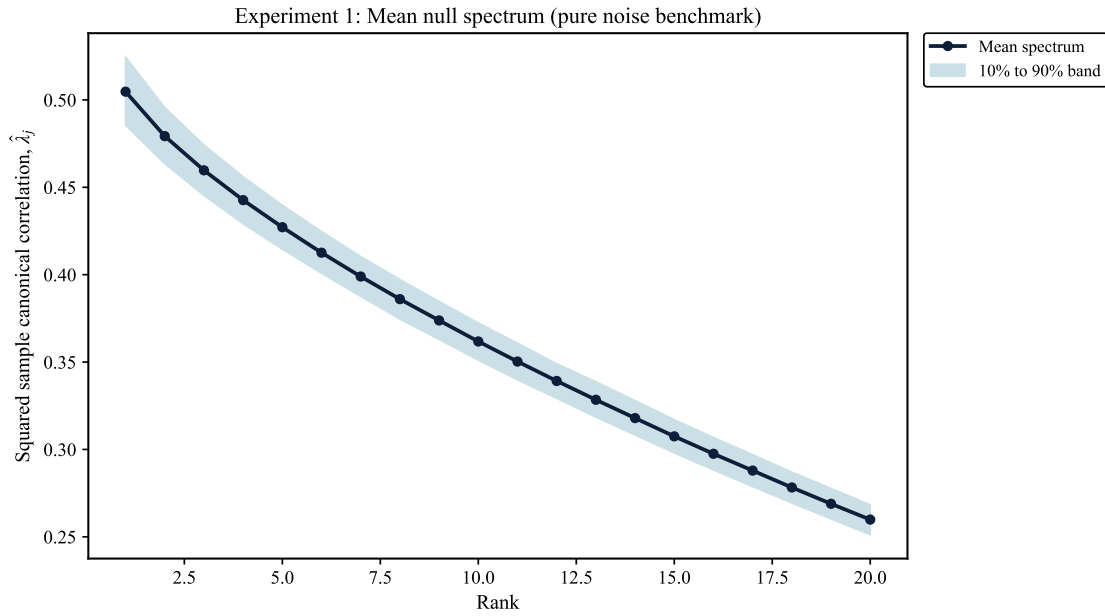
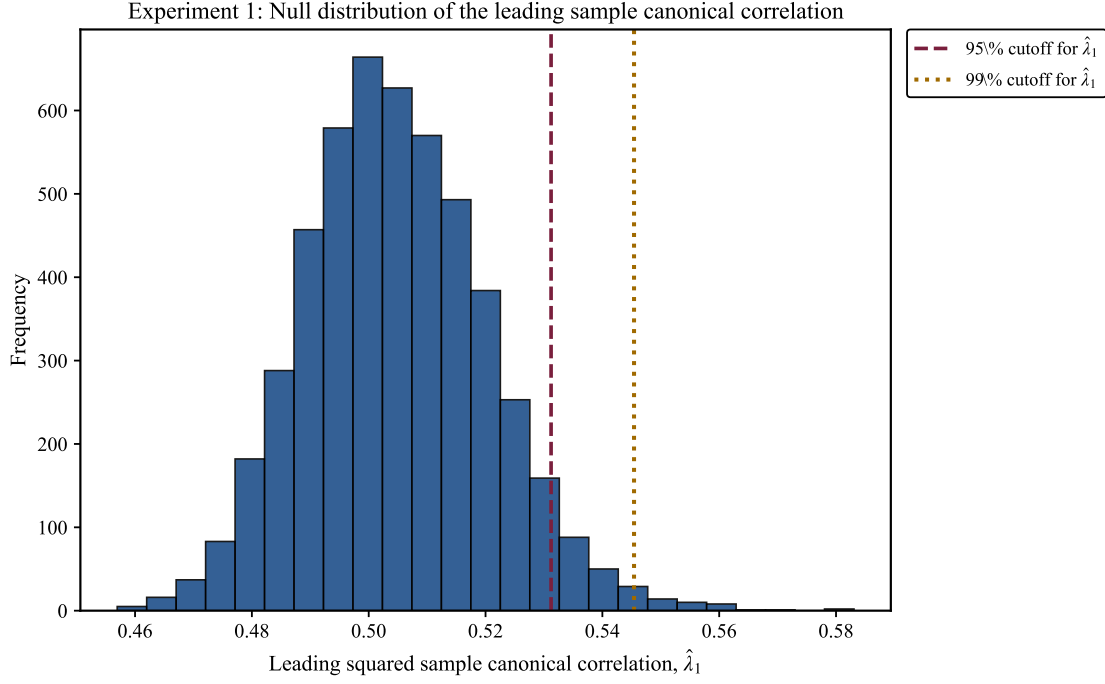


Figure 7.2*Experiment 1 null distribution of the leading sample canonical correlation.***Experiment 2: One-Spike Signal-Strength Sweep**

ρ	ρ^2	Mean $\hat{\lambda}_1$	Mean $\hat{\lambda}_2$	Mean out-of-sample $\hat{\lambda}_1$	Mean split stability, X	Mean split stability, Y	Mean angle, X (deg)	Mean angle, Y (deg)
0.0000	0.0000	0.5047	0.4793	0.0040	0.0125	0.0127	84.9259	85.0234
0.0864	0.0075	0.5051	0.4796	0.0038	0.0133	0.0128	84.6203	84.5861
0.1727	0.0298	0.5061	0.4801	0.0039	0.0121	0.0130	83.8949	83.7077
0.2591	0.0671	0.5068	0.4809	0.0040	0.0122	0.0114	81.9520	81.9692
0.3455	0.1193	0.5098	0.4840	0.0039	0.0124	0.0127	77.8259	77.7542
0.4318	0.1865	0.5183	0.4886	0.0039	0.0127	0.0132	67.8541	68.0116
0.5182	0.2685	0.5410	0.4944	0.0052	0.0145	0.0137	52.4513	52.4379
0.6045	0.3655	0.5864	0.4976	0.0093	0.0184	0.0187	38.4585	38.3813
0.6909	0.4774	0.6509	0.4996	0.0358	0.0532	0.0528	29.3479	29.3770
0.7773	0.6042	0.7309	0.4994	0.1410	0.2130	0.2144	22.2394	22.1676
0.8636	0.7459	0.8257	0.4996	0.3965	0.5320	0.5300	15.8118	15.8590
0.9500	0.9025	0.9325	0.5001	0.7541	0.8347	0.8349	8.8506	8.8809

Table 7.2*Experiment 2 summary table for the one-spike signal-strength sweep.*

Figure 7.3

Experiment 2 in-sample spectral response to one true spike.

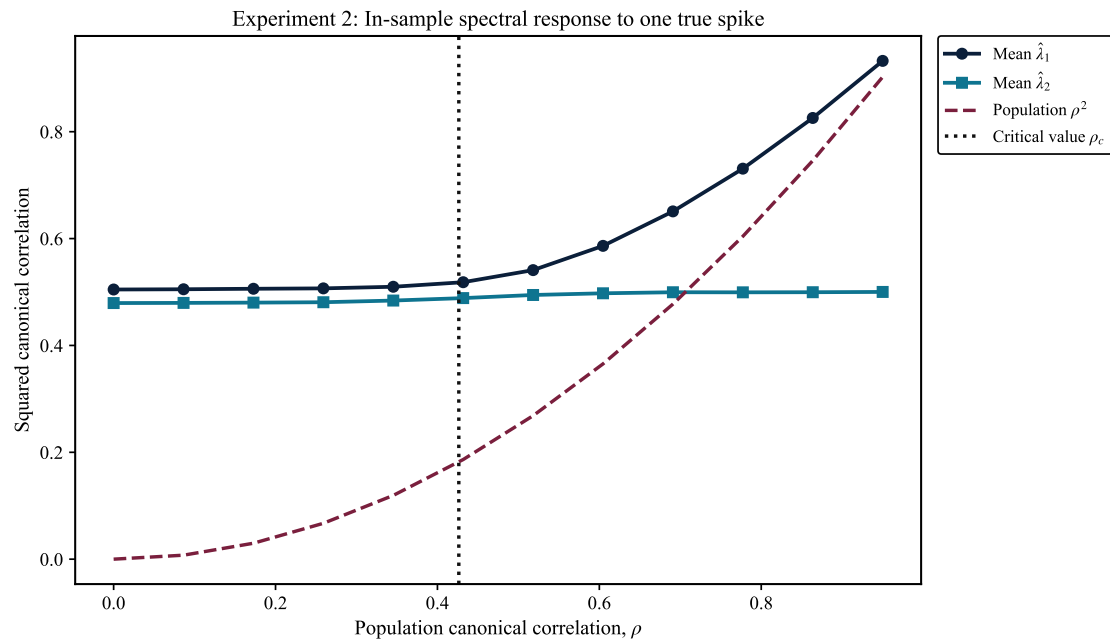


Figure 7.4

Experiment 2 out-of-sample strength and split-sample stability.

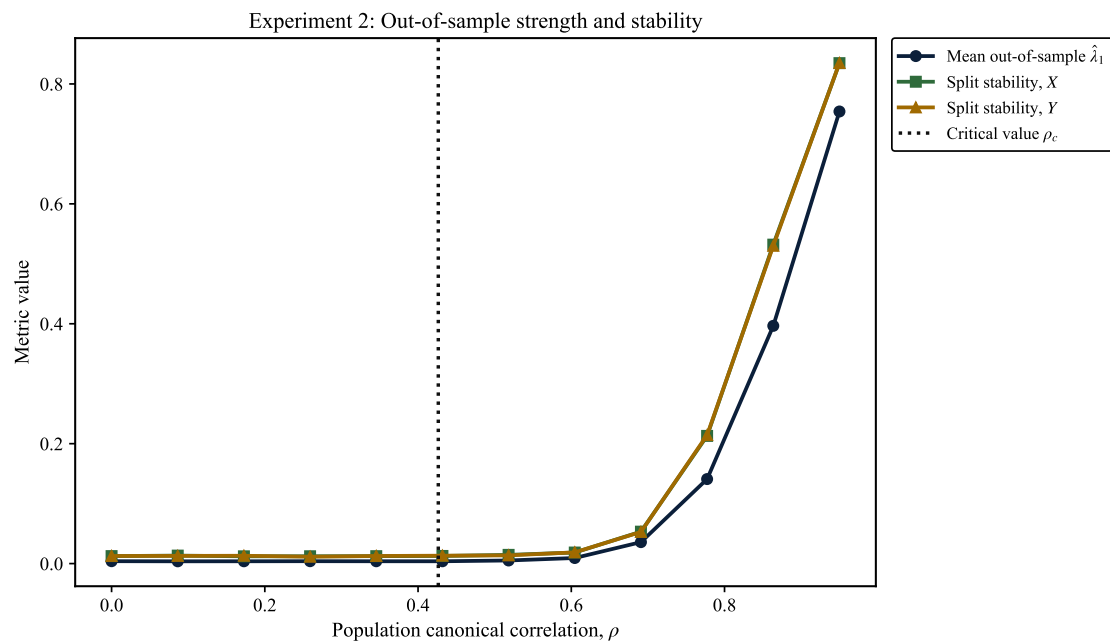
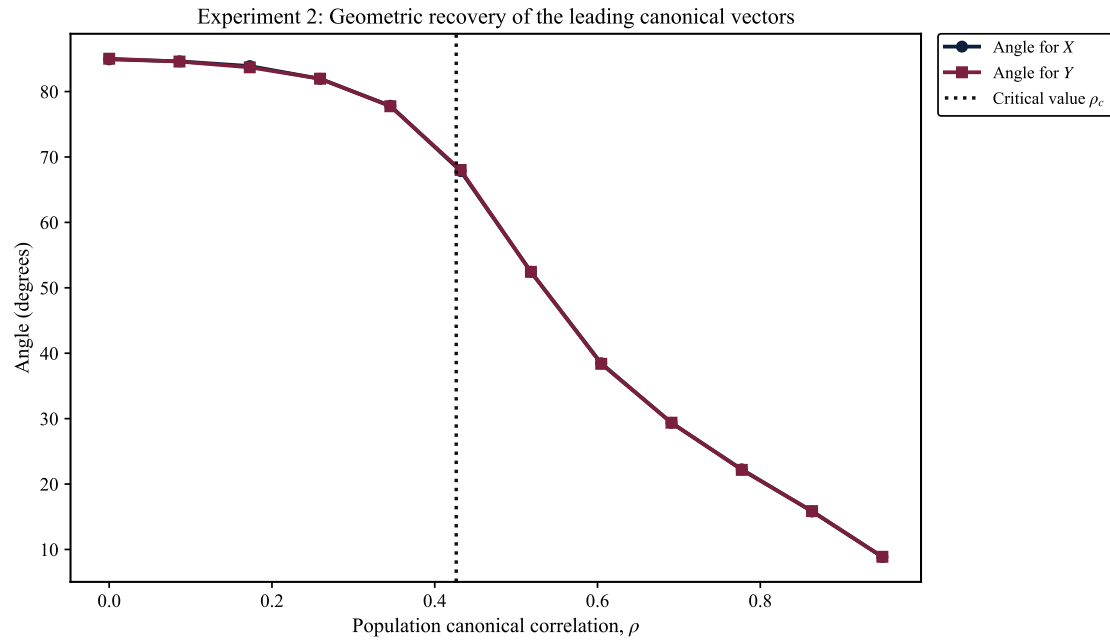


Figure 7.5*Experiment 2 geometric recovery of the leading canonical vectors.***Experiment 3: Threshold Zoom**

ρ	Detection probability	Mean $\hat{\lambda}_1$	Mean out-of-sample $\hat{\lambda}_1$	Mean split stability, X	Mean split stability, Y	Mean angle, X (deg)	Mean angle, Y (deg)
0.3064	0.0148	0.5087	0.0038	0.0125	0.0124	79.9697	79.9587
0.3464	0.0190	0.5099	0.0038	0.0131	0.0122	77.5343	77.5434
0.3864	0.0326	0.5129	0.0040	0.0128	0.0121	73.6238	73.6935
0.4264	0.0598	0.5171	0.0040	0.0126	0.0125	69.0789	69.1395
0.4664	0.1370	0.5245	0.0044	0.0131	0.0134	62.6661	62.8044
0.5064	0.3324	0.5365	0.0050	0.0139	0.0135	54.6415	54.7019
0.5464	0.6130	0.5531	0.0058	0.0145	0.0144	47.1378	47.1543

Table 7.3*Experiment 3 summary table around the near-critical region.*

Figure 7.6

Experiment 3 detection probability around the critical threshold.

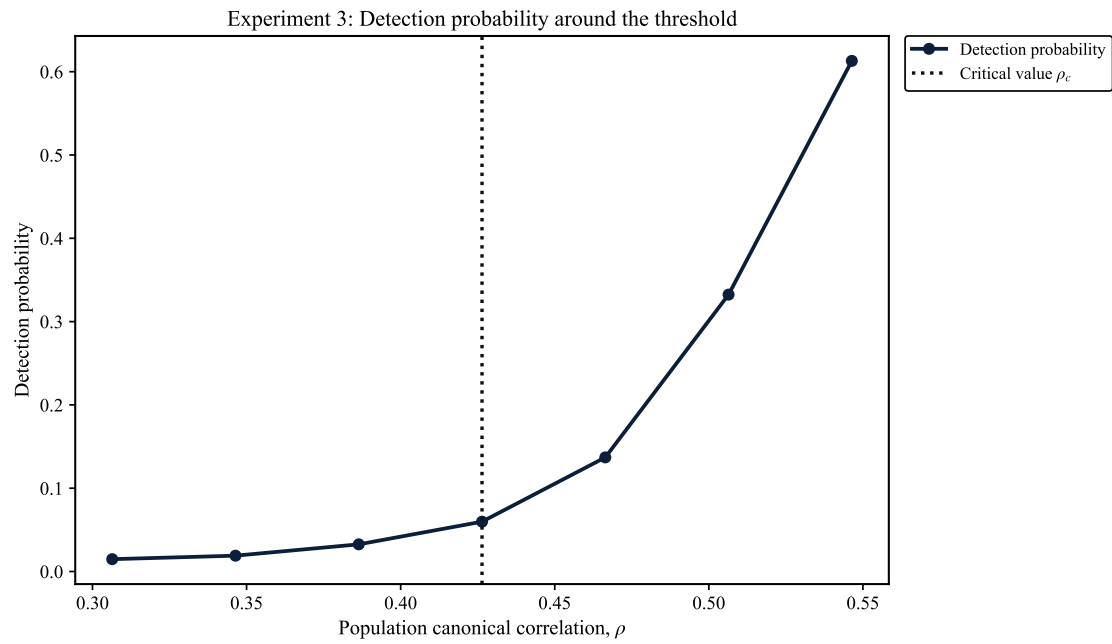


Figure 7.7

Experiment 3 in-sample and out-of-sample strength near the threshold.

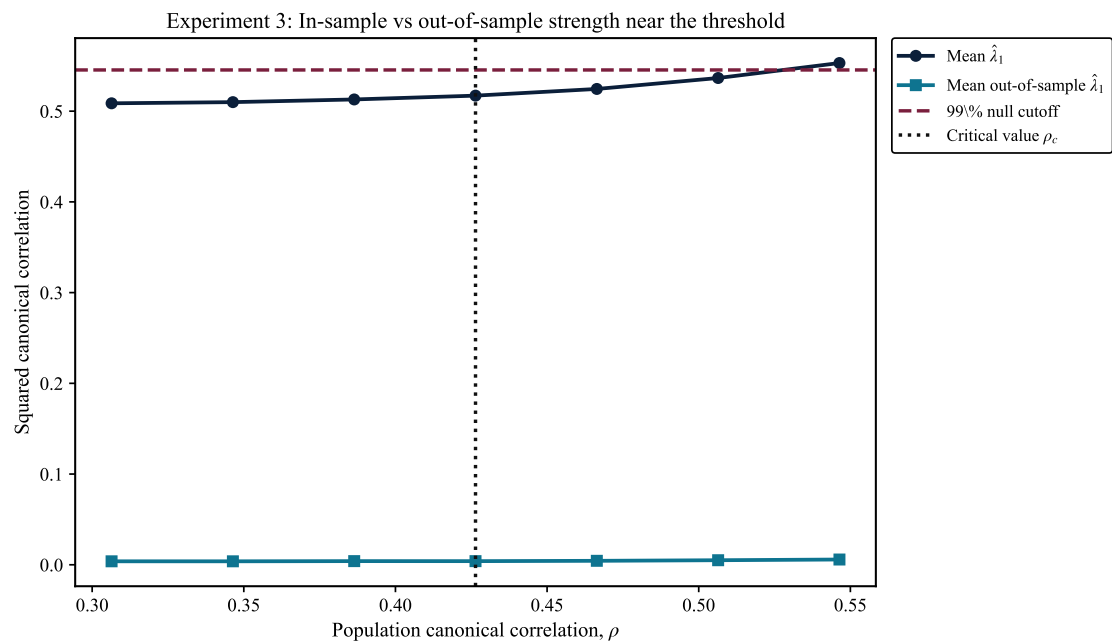
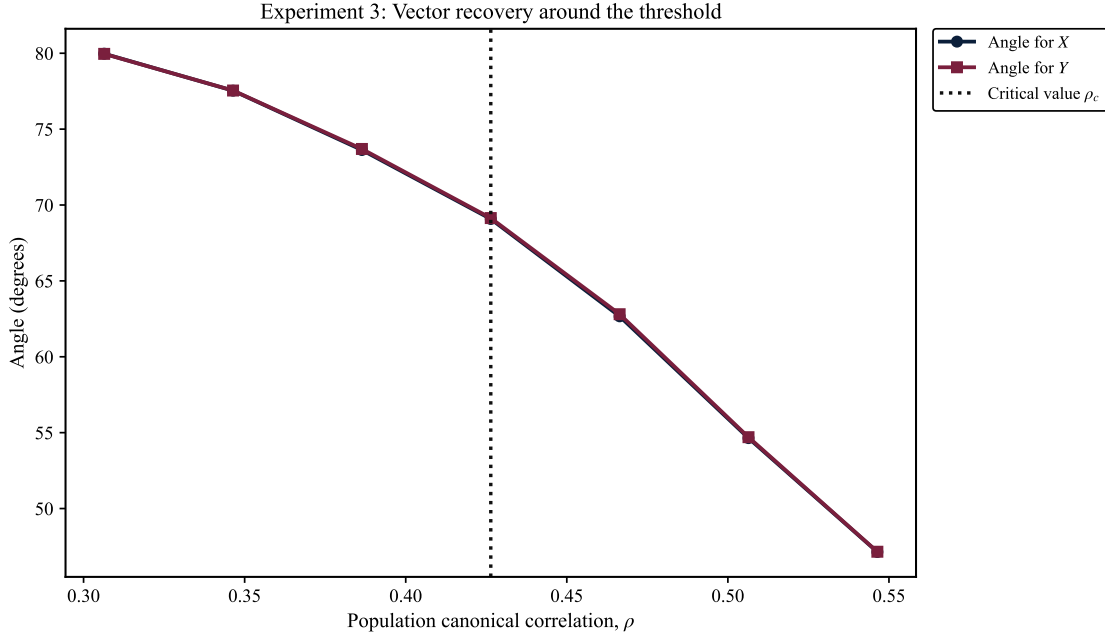


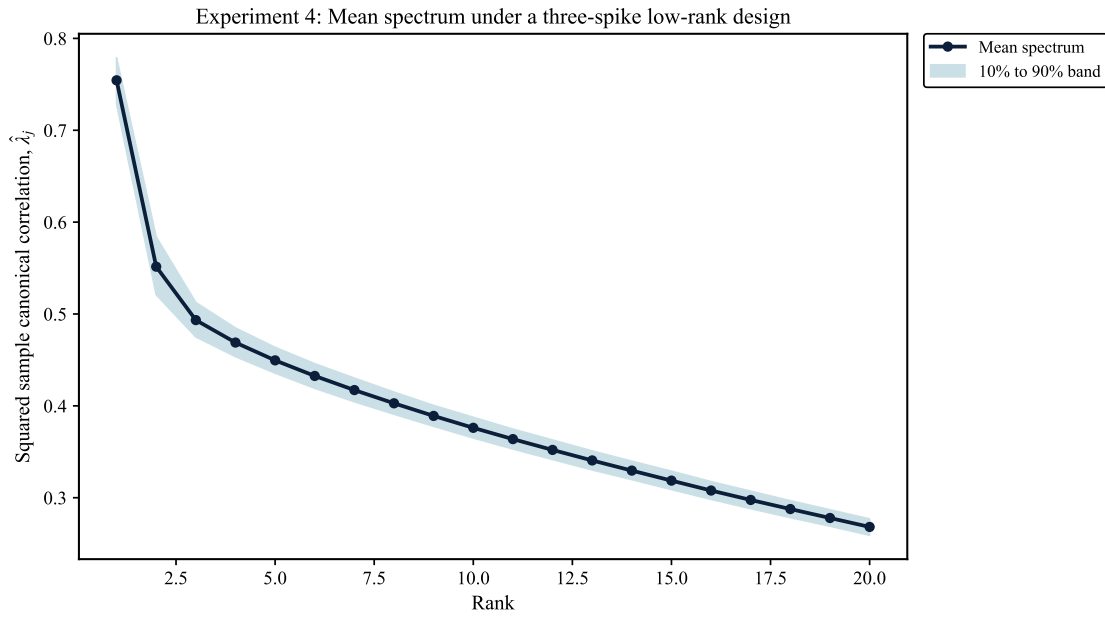
Figure 7.8*Experiment 3 vector recovery around the threshold.***Experiment 4: Multiple-Spike Low-Rank Design**

Metric	Value
Mean number of outliers	1.5785
Mean out-of-sample $\hat{\lambda}_1$	0.1881
Mean split stability, X	0.2718
Mean split stability, Y	0.2715
Mean $\hat{\lambda}_1$	0.7544
Mean $\hat{\lambda}_2$	0.5515
Mean $\hat{\lambda}_3$	0.4934
Mean angle for component 1, X (deg)	20.8440
Mean angle for component 2, X (deg)	47.0528
Mean angle for component 3, X (deg)	80.4530

Table 7.4*Experiment 4 summary statistics for the three-spike low-rank design.*

Component	Mean $\hat{\lambda}_j$	Mean angle, X (deg)	Mean squared cosine, X
1	0.7544	20.8440	0.8725
2	0.5515	47.0528	0.4692
3	0.4934	80.4530	0.0407

Table 7.5*Experiment 4 component-by-component recovery measures.*

Figure 7.9*Experiment 4 mean scree plot under the three-spike low-rank design.***Experiment 5: Diffuse Versus Low-Rank Dependence**

Design	Signal rank	Signal energy, $\sum \rho_j^2$	Mean number of outliers	Mean $\hat{\lambda}_1$	Mean $\hat{\lambda}_2$	Mean $\hat{\lambda}_3$	Mean out-of-sample $\hat{\lambda}_1$	Mean split stability, X	Mean split stability, Y
Diffuse	10.0000	0.7620	0.1080	0.5269	0.5006	0.4801	0.0047	0.0128	0.0120
Low-rank	3.0000	1.2428	2.1100	0.7350	0.5943	0.5219	0.1417	0.1886	0.1893

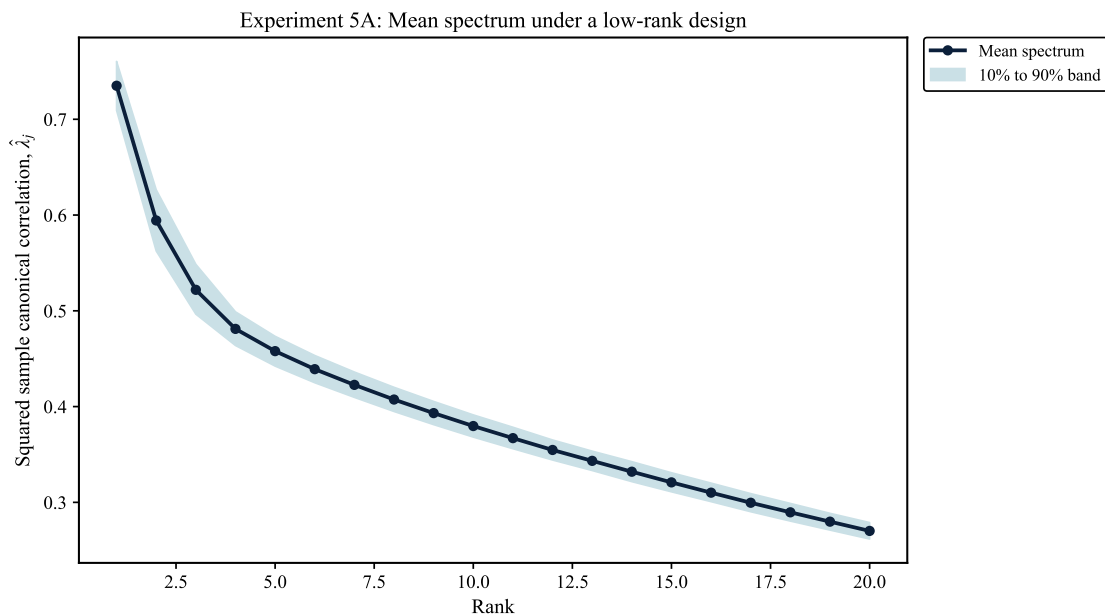
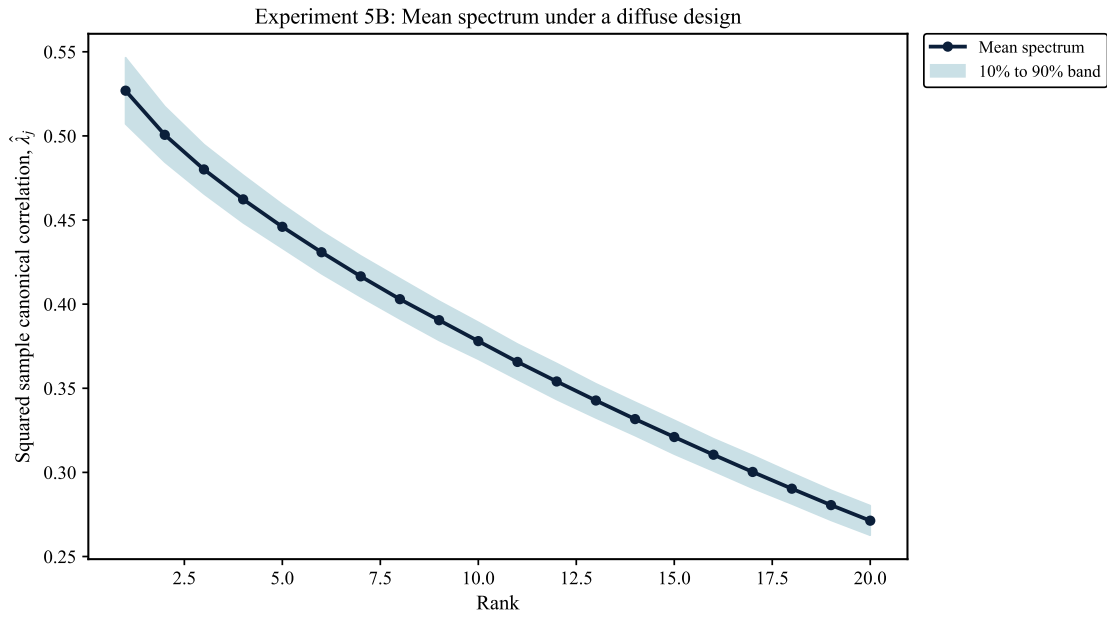
Table 7.6*Experiment 5 comparison of the low-rank and diffuse designs.***Figure 7.10***Experiment 5A mean scree plot under the low-rank design.*

Figure 7.11
Experiment 5B mean scree plot under the diffuse design.

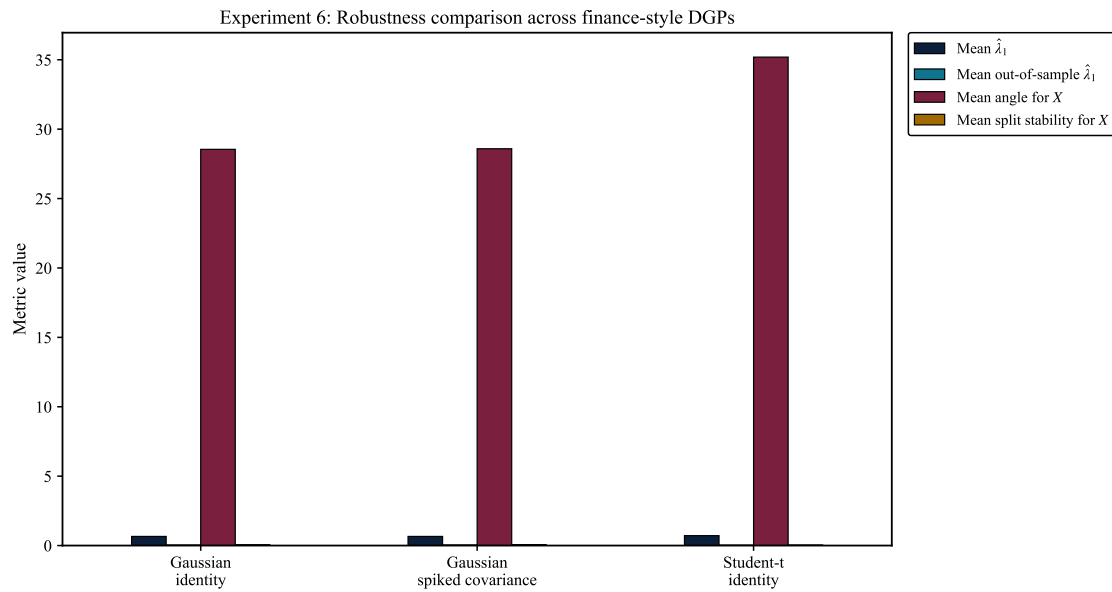


Experiment 6: Finance-Style Robustness Checks

Setting	Mean $\hat{\lambda}_1$	Mean out-of-sample $\hat{\lambda}_1$	Detection probability	Mean split stability, X	Mean split stability, Y	Mean angle, X (deg)	Mean angle, Y (deg)	Mean squared cosine, X	Mean squared cosine, Y
Gaussian identity	0.6578	0.0394	1.0000	0.0591	0.0583	28.5458	28.5406	0.7702	0.7704
Gaussian spiked covariance	0.6577	0.0417	1.0000	0.0618	0.0605	28.5892	28.5546	0.7697	0.7701
Student-t identity	0.7135	0.0314	1.0000	0.0403	0.0409	35.1913	35.2037	0.6671	0.6667

Table 7.7
Experiment 6 summary statistics across the robustness designs.

Figure 7.12
Experiment 6 robustness comparison across finance-style data-generating processes.



7.2. APPENDIX 2: Empirical Application Supplementary Results

This appendix collects the supplementary tables and figures for the CRSP empirical application in Chapter 5. Table 7.8 reports the baseline sample design used in the rolling analysis. Table 7.9 and Figure 7.13 summarize the full-sample static spectrum, which confirms that the presence of multiple outliers is not solely a rolling-window artifact.

Item	Value
Sample span	2010-01-08 to 2025-12-12
Total weekly observations	832
Cyclical assets (p)	80
Defensive assets (q)	80
Rolling window length	260 weeks
Step size	4 weeks
Number of windows	144
Window bulk edge λ_+	0.8521
Critical population threshold ρ_c^2	0.4444
Coverage threshold	0.99

Table 7.8

Baseline CRSP sample design used in the empirical rolling-window application.

Rank	Squared root	Above bulk edge
1	0.9389	Yes
2	0.7588	Yes
3	0.6457	Yes
4	0.5956	Yes
5	0.4834	Yes
6	0.4583	Yes
7	0.4191	Yes
8	0.3920	Yes
9	0.3754	Yes
10	0.3606	Yes

Table 7.9

Top ten squared canonical roots from the full-sample static spectrum, with the HDCCA bulk-edge classification.

Figure 7.13

Full-sample static canonical spectrum for the CRSP cyclical and defensive panels.

